Final Review Cheat Sheet

For your eyes only during the demo/Q&A - not a slide. Automated Data Health & Profiling Dashboard. Updated 2026-09-26.

Demo Commands (run in the project folder)

```
python cli.py sample_200.csv (main demo: both methods + comparison)
python cli.py sample_200.csv --method knn --k 7
python cli.py sample_200.csv --method isolation_forest
python cli.py sample_messy.csv --contamination 0.15 (robustness: mixed types, blanks, ragged row)
python cli.py sample_numeric.csv --no-impute (shows the skip-incomplete-rows behaviour)
```

If the live demo fails: narrate over the sample-output slide instead.

Anticipated Questions

Q	A
Why two methods instead of one?	They look for different things: Isolation Forest asks how easily a row is cut off; KNN asks how far it is from its nearest neighbours. Rows flagged by both are high-confidence; rows flagged by one are worth a look.
How well did each do on the planted outliers?	sample_200.csv has 8 planted outliers; each method flags 10 rows. KNN found 8/8, Isolation Forest 7/8 (missed id 145). At contamination 0.10 both find 8/8, with 20 rows flagged.
Why did Isolation Forest miss id 145?	Random blanks wiped out its income and score, so median imputation hid what made it unusual. KNN still caught it through age and tenure. It is a known trade-off of imputing before scoring.
Why does KNN standardise the columns but Isolation Forest doesn't?	KNN uses distances, which are meaningless if income (tens of thousands) and age (tens) aren't on the same scale. Isolation Forest splits each column separately, so scale matters much less.
Why is 8/8 for KNN not proof that KNN is better?	One controlled test on made-up data with known answers, at one setting. On real data there is no answer key. The two methods complement each other.
What does contamination control?	The expected share of outliers; both methods flag that top fraction (default 0.05). Adjustable with --contamination (0 to 0.5).
Why median imputation instead of mean?	The median isn't dragged around by the extreme values we are about to search for. Statistics are computed on the raw data; filling happens only in a working copy for scoring.
What about text/categorical columns?	Excluded from statistics and scoring for now; encoding is listed as future work.
Why CLI before Streamlit?	Scoped terminal-only for the deadline. The modules are pure functions so a Streamlit UI can call them directly without a rewrite.

Key Numbers To Have Ready

- sample_200.csv: 200 rows, 8 planted outliers, 10 flagged per method. Isolation Forest 7/8, KNN 8/8; both agree on 8 rows (12 distinct rows flagged); score rank correlation 0.86.
- Defaults: --method both, --contamination 0.05, --k 5, median imputation on (--no-impute to disable).
- Test fixtures: sample_numeric.csv, sample_messy.csv, sample_fully_missing_col.csv, sample_200.csv.